Customer Churn Prediction Using Machine Learning

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Abstract

This project investigates customer churn prediction in the telecom industry using Logistic Regression. The dataset, sourced from Kaggle, contains over 500,000 customer records with features including support call frequency, payment delay, contract length, and total spend. Performance is assessed using overall accuracy and recall for the non-churned class on a held-out test set.

Initial results reveal a significant discrepancy between training accuracy (89.5%) and test accuracy (57.1%), with a recall of only 19.4% for the non-churned class. Through distribution analysis, we identify a substantial distribution shift in the original train/test split, particularly in the Not Churn group. Following a random re-split of the dataset, the model achieves a test accuracy of 84.7% and a recall of 84.7% for the non-churned class, demonstrating that the original performance degradation was attributable to data partitioning rather than model limitations.

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1 Introduction

Customer churn refers to the rate at which customers discontinue their subscription or cease using a company's products and services within a given period. For telecom companies, churn is a critical business metric, as retaining existing customers is substantially more cost-effective than acquiring new ones. Identifying customers at risk of churning enables companies to intervene proactively, through targeted retention strategies, personalized offers, or service improvements, before the customer decides to leave.

This project investigates customer churn prediction using Logistic Regression on a telecom customer dataset of over 500,000 records. Rather than focusing solely on maximizing classifier performance, this work emphasizes identifying and addressing data quality issues that affect model reliability. Through exploratory data analysis and model evaluation, we uncover a distribution shift in the original train/test split, analyze its cause, and demonstrate that correcting the data partitioning substantially improves generalization performance.

2 Data

2.1 Dataset Overview

The dataset contains 505,207 customer records sourced from Kaggle, representing telecom subscribers over a 12-month observation period. The dataset is pre-divided into a training set (440,833 observations) and a test set (64,374 observations). During preprocessing, one incomplete observation was removed from the training set, resulting in a final training set of 440,832 records. The target variable *Churn* is binary, where 1 indicates a customer who discontinued their subscription and 0 indicates a retained customer. The training set contains 190,833 non-churned (43.3%) and 249,999 churned (56.7%) customers, indicating a mild class imbalance.

2.2 Feature Description

The dataset contains 11 variables: 1 identifier (CustomerID, excluded from analysis), 1 binary target (Churn), and 10 predictors. Features span demographic (Age, Gender), behavioral (Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction), and contractual (Tenure, Subscription Type, Contract Length) dimensions. Among the 10

predictors, 3 are categorical and 7 are continuous.

3 Exploratory Data Analysis

3.1 Summary Statistics

All continuous features fall within expected ranges, with no extreme outliers identified. One incomplete observation was detected in the training set and removed prior to analysis, resulting in a final training set of 440,832 records.

3.2 Feature Distribution by Churn

Figure 1 shows the overall churn distribution in the training set. Of 440,832 observations, 190,833 (43.3%) are non-churned and 249,999 (56.7%) are churned, indicating a mild class imbalance that is unlikely to substantially affect model performance.

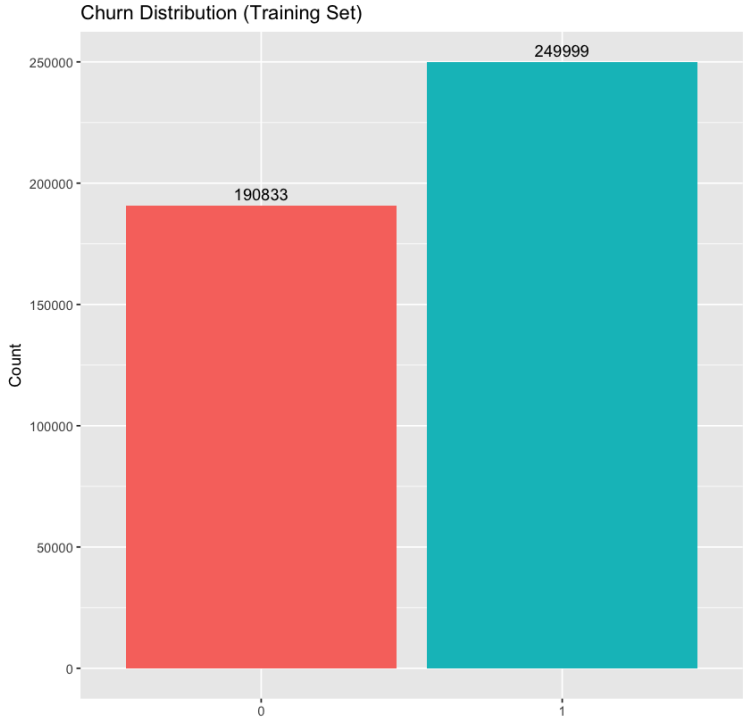

Figure 1: Churn Distribution (Training Set)

Figure 2 shows the churn rate across categorical features. Contract Length shows the most

notable pattern: customers on Monthly contracts exhibit a substantially higher churn rate compared to those on Quarterly or Annual plans. This suggests that the absence of a long-term commitment is strongly associated with churn risk. Gender and Subscription Type show relatively balanced churn distributions across categories, indicating these features may have limited discriminative power on their own.

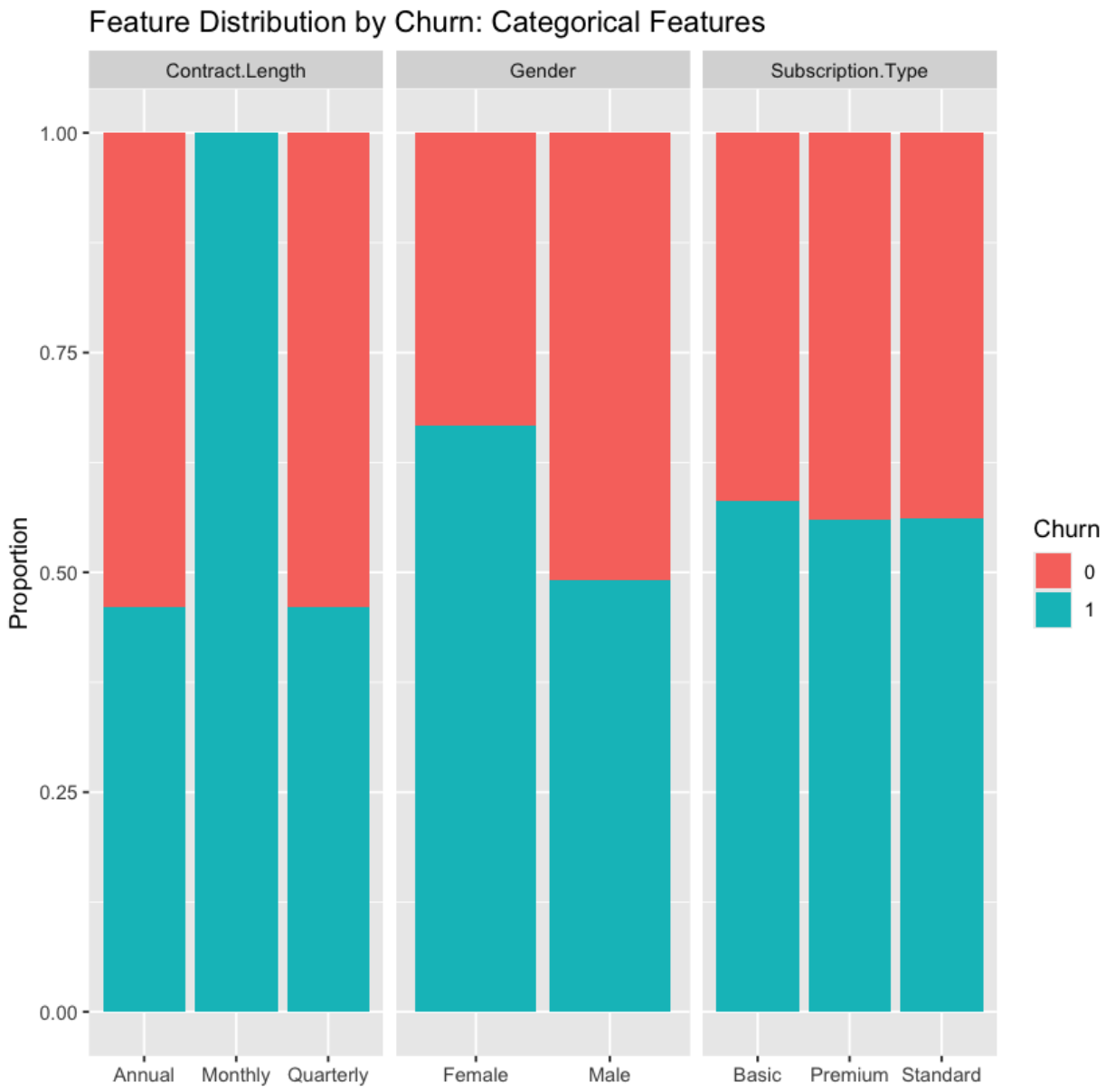

Figure 2: Feature Distribution by Churn: Categorical Features

Figure 3 shows the distribution of numeric features by churn status. Support Calls shows the most pronounced difference between churned and non-churned customers, with churned

customers making substantially more calls. This suggests that higher support call frequency reflects customer dissatisfaction and is a strong indicator of churn risk. Other features such as Tenure and Usage Frequency show minimal differences between the two groups.

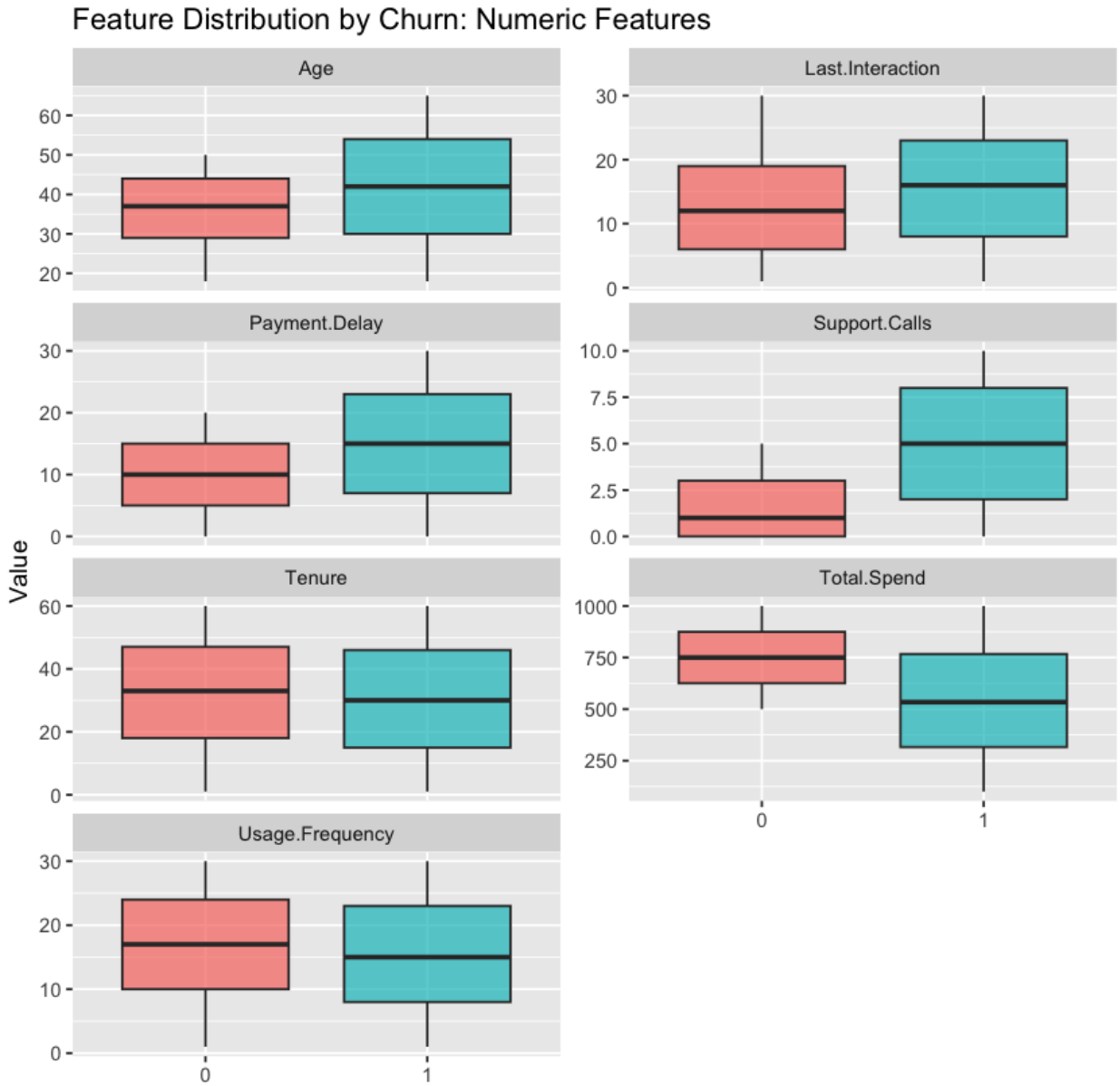

Figure 3: Feature Distribution by Churn: Numeric Features

3.3 Feature Correlation

Support Calls shows the strongest positive correlation with Churn ($r = 0.57$), consistent with the pattern observed in Section 3.2, confirming that higher support call frequency is a strong indicator of churn risk. Total Spend shows the strongest negative correlation ($r = -0.43$), suggesting that customers with higher spending are less likely to leave. This is intuitive: loyal, high-value customers tend to be more engaged with the service, while customers who are already considering leaving may have reduced their usage and spending before eventually churning. The remaining features show weak to moderate correlations with Churn.

3.4 Multicollinearity Check (VIF)

4 Methodology

4.1 Data Preprocessing

4.2 Feature Selection (Backward Elimination)

4.3 Logistic Regression

5 Results

5.1 Logistic Regression (Original Split)

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6 Discussion & Conclusion

6.1 Key Findings

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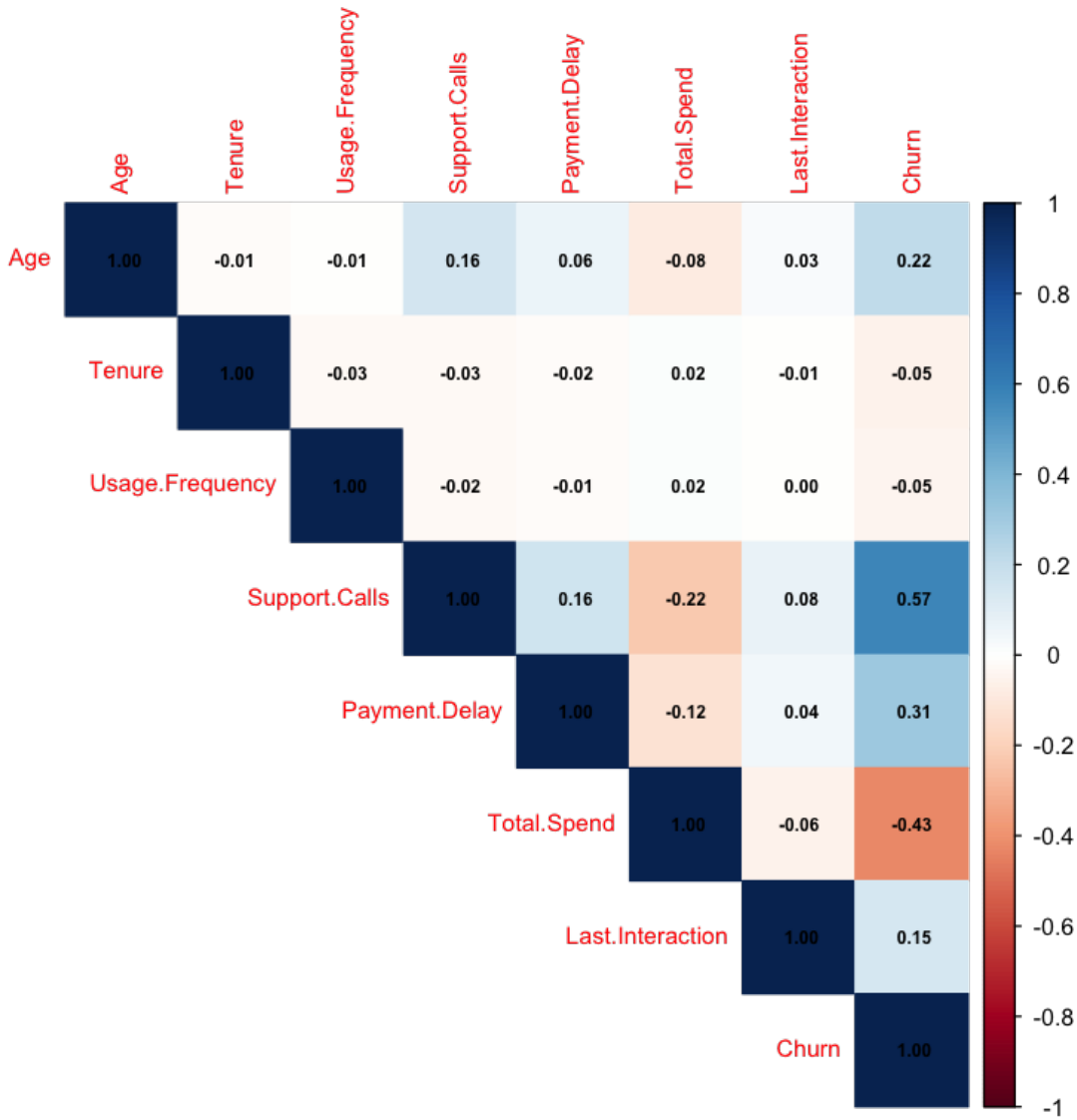

Figure 4: Feature Correlation Matrix